Unit – 3
Mixed Reality:
Difference between AR VR and MR

AR vs VR vs MR:

Parameter	Augmented Reality (AR)	Virtual Reality (VR)	Mixed Reality (MR)
Definition	AR **adds digital elements** (images, text, 3D objects) on top of the real world.	VR creates a completely **computer-generated virtual environment** that replaces the real world.	MR **blends real and virtual worlds**, allowing digital and physical objects to interact with each other.
Real World Visibility	The real world is **visible with digital overlays**.	The real world is **completely blocked**.	The real world is **visible and interactive with digital objects**.
User Experience	Semi-immersive experience where digital information is shown over the real environment.	Fully immersive experience inside a **virtual environment**	Highly immersive experience where **virtual objects interact with the real environment**.

AR vs VR vs MR:

Parameter	Augmented Reality (AR)	Virtual Reality (VR)	Mixed Reality (MR)
Devices	Smartphones, tablets, AR glasses.	Opaque headsets (e.g., Meta Quest, HTC Vive).	Advanced transparent headsets (e.g., HoloLens).
Immersion	Low; user remains fully aware of surroundings.	High; user is completely isolated from reality.	High; partial immersion with a sense of presence.
Hardware Required	Usually works with **smartphones, tablets, or AR glasses**.	Requires **VR headsets** and controllers.	Requires **advanced MR headsets** with sensors and cameras.
Environment Type	**Real environment enhanced** with digital content.	Completely **simulated environment**.	**Hybrid environment** combining real and virtual worlds.

AR vs VR vs MR:

Parameter	Augmented Reality (AR)	Virtual Reality (VR)	Mixed Reality (MR)
Interaction Level	Limited interaction with digital elements.	Interaction happens only with **virtual objects**.	**Advanced interaction** between real and virtual objects.
Examples	Pokémon Go, Snapchat filters.	Virtual gaming worlds, flight simulators.	Interactive holograms used in engineering or medical training.
Applications	Retail, navigation, education, marketing.	Gaming, training simulations, virtual tours, education.	Industrial design, medical visualization, advanced training.

Thank You…!!!